Pauli Matrix Formalism and Lie Groups

Structure of $SL(2,\mathbb{C})$ and $SU(2)$

Theoretical Physics Master - Group Theory

Master Study Guide

Overview

Objective: To develop the Pauli matrix function formalism to rigorously characterize the structure, algebra, and topology of the main Lie groups.

Definition of Pauli Matrix Function

Definition (9.1.1)

A Pauli matrix function is an element $\sigma \in \mathrm{Hom}(\mathbb{C}^3, \mathbb{C}[2])$ satisfying the relations

$$\sigma(x)^+ = \sigma(x^*)$$
$$\sigma(x)\sigma(y) = x \cdot y 1_2 + i\sigma(x \times y)$$

for all $x, y \in \mathbb{C}^3$.

Algebraic Properties I

Proposition (9.1.1)

One has

$$\sigma(x)\sigma(y) + \sigma(y)\sigma(x) = 2x \cdot y 1_2,$$
$$\sigma(x)\sigma(y) - \sigma(y)\sigma(x) = 2i\sigma(x \times y)$$

for all $x, y \in \mathbb{C}^3$.

Algebraic Properties II (Traces)

Proposition (9.1.2)

One has

$$\begin{aligned} \mathrm{tr}(\sigma(x)) &= 0 \\ \mathrm{tr}(\sigma(x)\sigma(y)) &= 2x \cdot y \\ \mathrm{tr}(\sigma(x)\sigma(y)\sigma(z)) &= 2ix \times y \cdot z \end{aligned}$$

for all $x, y, z \in \mathbb{C}^3$.

Algebraic Properties III (Determinant)

Proposition (9.1.3)

One has

$$\det \sigma(x) = -x^2$$

for $x \in \mathbb{C}^3$.

Expansion Theorem

Proposition (9.1.4)

For $X \in \mathbb{C}[2]$, there exist unique $x_0 \in \mathbb{C}$, $x \in \mathbb{C}^3$ such that

$$X = x_0 1_2 + \sigma(x)$$

Proposition (9.1.5)

Let $X \in \mathbb{C}[2]$ be a matrix expanded as above. Then,

$$\operatorname{tr} X = 2x_0$$

Determinant and Inverse

Proposition (9.1.6)

Let $X \in \mathbb{C}[2]$ be a matrix expanded as $X = x_0 1_2 + \sigma(x)$. Then,

$$\det X = x_0^2 - x^2$$

Proposition (9.1.7)

Let $X \in \mathbb{C}[2]$ be a matrix expanded in the form $X = x_0 1_2 + \sigma(x)$ with $x_0^2 - x^2 \neq 0$. Then, X is invertible and

$$X^{-1} = \frac{1}{x_0^2 - x^2}(x_0 1_2 - \sigma(x))$$

Characteristic Reflection

Proposition (9.2.1)

There exists a matrix $\Pi \in \mathbb{R}[3]$ such that

$$\Pi = \Pi^t$$
$$\Pi^2 = 1$$
$$\det \Pi = -1$$

and with the property that $\sigma(x)^t = \sigma(\Pi x)$ and $\sigma(x)^* = \sigma(\Pi x^*)$ for $x \in \mathbb{C}^3$.

Characteristic Projectors

Definition (9.2.1)

We let $P_{\pm 1} \in \mathbb{R}[3]$ be the matrices

$$P_{\pm 1} = \frac{1}{2}(1 \pm \Pi)$$

Proposition (9.2.2)

The characteristic projectors $P_{\pm 1}$ satisfy:

$$\begin{aligned} P_{\pm 1}^2 = P_{\pm 1}, \quad P_{\pm 1} P_{\mp 1} = 0, \quad P_{+1} + P_{-1} = 1 \\ \Pi = P_{+1} - P_{-1}, \quad \operatorname{tr} P_{+1} = 2, \quad \operatorname{tr} P_{-1} = 1 \end{aligned}$$

Characteristic Eigenspaces

Definition (9.2.2)

We let $E_{\pm 1} \subset \mathbb{R}^3$ be the eigenspaces of Π relative to the eigenvalues ± 1, respectively.

Proposition (9.2.3 & 9.2.4)

The dimensions are $\dim E_{+1} = 2$ *and* $\dim E_{-1} = 1$*.*
Let $x \in \mathbb{R}^3$*. Then, one has*

$$\sigma(x)^t = \sigma(x)^* = \pm\sigma(x)$$

if and only if $x \in E_{\pm 1}$*.*

Characteristic Unit Vector

Proposition (9.2.5)

There is a unique vector $n \in \mathbb{R}^3$ *with* $|n| = 1$ *such that*

$$\sigma(n) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Further, $E_{-1} = \mathbb{R}n$.

The Standard Pauli Matrices

Definition (9.3.1)

The Pauli matrices $\sigma_i \in \mathbb{C}[2]$, $i = 1, 2, 3$ are given by

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Proposition (9.3.1)

For $i, j = 1, 2, 3$, one has

$$\sigma_i^+ = \sigma_i$$

$$\sigma_i \sigma_j = \delta_{ij} 1_2 + i \sum_{k=1}^{3} \epsilon_{ijk} \sigma_k$$

The Standard Pauli Matrix Function

Definition (9.3.2)

Define $\sigma : \mathbb{C}^3 \to \mathbb{C}[2]$ by $\sigma(x) = \sum_{i=1}^{3} x_i \sigma_i$ with $x \in \mathbb{C}^3$. Explicitly:

$$\sigma(x) = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}$$

Proposition (9.3.2 & 9.3.3)

σ is a Pauli matrix function.
Let σ be a Pauli matrix function and $U \in U(2)$ a unitary matrix. Then $\sigma_U(x) = U\sigma(x)U^+$ is a Pauli matrix function as well.

The Exponential of a Pauli Matrix Function

Proposition (9.4.1)

One has, for $x \in \mathbb{C}^3$

$$\exp\left(\frac{1}{2}\sigma(x)\right) = \cosh\left(\frac{l(x)}{2}\right) 1_2 + \frac{1}{l(x)} \sinh\left(\frac{l(x)}{2}\right) \sigma(x)$$

where the complex length $l(x)$ of x is defined by $l(x) = (x^2)^{1/2}$.

Properties of the Exponential

Proposition (9.4.2 & 9.4.3)

For $x \in \mathbb{C}^3$:

$$\operatorname{tr}\exp\left(\frac{1}{2}\sigma(x)\right) = 2\cosh\left(\frac{I(x)}{2}\right)$$

$$\det\exp\left(\frac{1}{2}\sigma(x)\right) = 1$$

Injectivity of the Exponential

Proposition (9.4.4)

Let $x, y \in \mathbb{C}^3$ such that

$$\exp\left(\frac{1}{2}\sigma(x)\right) = \exp\left(\frac{1}{2}\sigma(y)\right)$$

$$|l(x) \pm l(y)| < 4\pi \quad \textit{(with both signs)}$$

Then, $x = y$.

The Group $SL(2,\mathbb{C})$

Proposition (10.1.1)

For any $A \in SL(2,\mathbb{C})$, there exist unique $w \in \mathbb{C}$, $a \in \mathbb{C}^3$ with $w^2 - a^2 = 1$ such that

$$A = w1_2 + \sigma(a)$$

Viceversa, if $A \in \mathbb{C}[2]$ is of the above form, then $A \in SL(2,\mathbb{C})$.

Definition (10.1.1)

For $w \in \mathbb{C}$, $a \in \mathbb{C}^3$ with $w^2 - a^2 = 1$, let $A(w,a) \in SL(2,\mathbb{C})$,

$$A(w,a) = w1_2 + \sigma(a)$$

The Lie Algebra $\mathfrak{sl}(2,\mathbb{C})$

Proposition (10.1.2)

For any $X \in \mathfrak{sl}(2,\mathbb{C})$, there exists a unique $x \in \mathbb{C}^3$ such that $X = \sigma(x)$. Viceversa, if $X \in \mathbb{C}[2]$ is of the above form, then $X \in \mathfrak{sl}(2,\mathbb{C})$.

Definition (10.1.2)

For $x \in \mathbb{C}^3$, let $J(x) \in \mathfrak{sl}(2,\mathbb{C})$

$$J(x) = \sigma(x)/2$$

Lie Bracket and Generators of $\mathfrak{sl}(2,\mathbb{C})$

Proposition (10.1.3)

For $x, y \in \mathbb{C}^3$,

$$[J(x), J(y)] = iJ(x \times y)$$

Proposition (10.1.4)

Suppose that v_k, $k = 1, 2, 3$ is an oriented orthonormal basis of $\mathbb{R}^3$ regarded as a triple of vectors of $\mathbb{C}^3$. Then, $J(v_k)$ is a set of generators of $\mathfrak{sl}(2,\mathbb{C})$ and

$$[J(v_i), J(v_j)] = i\sum_{k=1}^{3} \epsilon_{ijk} J(v_k)$$

Exponential Map of $SL(2,\mathbb{C})$

Proposition (10.1.5)

For $x \in \mathbb{C}^3$, one has

$$\exp(J(x)) = \cosh\left(\frac{(x^2)^{1/2}}{2}\right) 1_2 + \frac{1}{(x^2)^{1/2}} \sinh\left(\frac{(x^2)^{1/2}}{2}\right) \sigma(x)$$

Proposition (10.1.6)

Let $A \in SL(2,\mathbb{C})$ such that $(1_2 + A)^2 \neq 0$. Then, there is $x \in \mathbb{C}^3$ such that

$$A = \exp(J(x))$$

The Group $SU(2)$

Proposition (10.2.1)

For every $U \in SU(2)$, there exist unique $c \in \mathbb{R}$, $s \in \mathbb{R}^3$ with $c^2 + s^2 = 1$ such that

$$U = c1_2 - i\sigma(s)$$

Viceversa, if $U \in \mathbb{C}[2]$ is of the above form, then $U \in SU(2)$.

Definition (10.2.1)

For $c \in \mathbb{R}$, $s \in \mathbb{R}^3$ with $c^2 + s^2 = 1$, let $U(c,s) \in SU(2)$,

$$U(c,s) = c1_2 - i\sigma(s)$$

The Lie Algebra $\mathfrak{su}(2)$

Proposition (10.2.2)

For every $X \in \mathfrak{su}(2)$, there exists a unique $x \in \mathbb{R}^3$ such that

$$X = -i\sigma(x)$$

Viceversa, if $X \in \mathbb{C}[2]$ is of the above form, then $X \in \mathfrak{su}(2)$.

Definition (10.2.2)

For $x \in \mathbb{R}^3$, let $S(x) \in \mathfrak{su}(2)$

$$S(x) = -i\sigma(x)/2$$

Lie Bracket and Generators of $\mathfrak{su}(2)$

Proposition (10.2.3)

For $x, y \in \mathbb{R}^3$,

$$[S(x), S(y)] = S(x \times y)$$

Proposition (10.2.4)

Suppose that v_k, $k = 1, 2, 3$ is an oriented orthonormal basis of $\mathbb{R}^3$. Then, $S(v_k)$ is a set of generators of $\mathfrak{su}(2)$ and

$$[S(v_i), S(v_j)] = \sum_{k=1}^{3} \epsilon_{ijk} S(v_k)$$

Exponential Map of $SU(2)$

Proposition (10.2.5)

For $x \in \mathbb{R}^3$, one has

$$\exp(S(x)) = \cos\left(\frac{|x|}{2}\right) 1_2 - \frac{i}{|x|} \sin\left(\frac{|x|}{2}\right) \sigma(x)$$

Proposition (10.2.6)

The map

$$\exp : \{X | X = S(\phi), \phi \in \mathbb{R}^3, |\phi| \leq 2\pi\} \to SU(2)$$

is surjective. Further, the map

$$\exp : \{X | X = S(\phi), \phi \in \mathbb{R}^3, |\phi| < 2\pi\} \to SU(2) \setminus \{-1_2\}$$

Backup: Topology of Lie Groups

The Pauli formalism provides an immediate topological classification of the analyzed groups:

$SL(2,\mathbb{C})$ and the Hyperboloid

Every $A \in SL(2,\mathbb{C})$ is described by $w \in \mathbb{C}$ and $a \in \mathbb{C}^3$ such that $w^2 - a^2 = 1$. This defines a **3-dimensional complex hyperboloid** $Y^{1,2}$ embedded in $\mathbb{C}^4$.

$SU(2)$ and the Sphere

Every $U \in SU(2)$ is described by $c \in \mathbb{R}$ and $s \in \mathbb{R}^3$ such that $c^2 + s^2 = 1$. This defines a **3-dimensional sphere** S^3 embedded in $\mathbb{R}^4$.

Backup: The Group $SO(2)$

Definition (10.3.1)

Setting $J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, for $\gamma, \varsigma \in \mathbb{R}$ with $\gamma^2 + \varsigma^2 = 1$, we define $R(\gamma, \varsigma) \in SO(2)$ as:

$$R(\gamma, \varsigma) = \gamma 1_2 + \varsigma J_2$$

This condition ($\gamma^2 + \varsigma^2 = 1$) topologically identifies $SO(2)$ with the circle S^1.

Proposition (10.3.5)

The Lie algebra $\mathfrak{so}(2)$ is generated by $K(x) = xJ_2$. The exponential map is:

$$\exp(K(x)) = \cos x 1_2 + \sin x J_2$$

Backup: The Group $U(1)$

Proposition (10.4.1 & 10.4.2)

The Lie algebra $\mathfrak{u}(1)$ is defined as $\mathfrak{u}(1) = \{ix | x \in \mathbb{R}\}$.
Its exponential map is the well-known Euler's formula:

$$\exp(ix) = \cos x + i \sin x$$

Surjectivity and Invertibility

The map $\exp : \{i\varphi | -\pi \leq \varphi < \pi\} \to U(1)$ is surjective.
Excluding -1, the map is invertible in the open interval $(-\pi, \pi)$.